MILITARY SPECIFICATION

CLOTH, PLAIN WEAVE, NYLON; CLOTH, PLAIN WEAVE, POLYESTER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers medium weight cloth made of synthetic multifilament yarns, and prepared for coating.

1.2 Classification. The cloth shall be of the following types:

Type I	-	Polyester, 4 ounce
Type II	-	Nylon, 4.9 ounce

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-P-1133 - Packaging and Packing of Synthetic Fiber Fabrics

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

FSC 8305

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

PUBLICATION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D. C. 20580.)

3. REQUIREMENTS

3.1 Material.

* 3.1.1 Yarn.

3.1.1.1 Type I. The warp and filling yarns shall be polyester, bright, multifilament, polyethylene glycol terephalate.

3.1.1.2 Type II. The warp and filling yarns shall be bright, multifilament nylon.

3.2 Color. The cloth shall be of a natural color.

3.3 Physical requirements. The finished cloth shall conform to the requirements of table I when tested as specified in 4.4.

TABLE I. Physical requirements

	Weight oz./sq.yd. (minimum)	Yarns per inch (minimum)		Breaking strength, pounds (minimum)	
		Warp	Filling	Warp	Filling
Type I	4.0	32	32	200	200
Type II	4.0	22	22	275	275

* 3.3.1 Weave. The weave shall be plain.

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3.3.2 Width. The width shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvages.

* 3.4 Finish. The cloth shall be scoured and heat set. The cloth shall be heat set to meet the requirements cited in 3.4.1.

3.4.1 Dimensional stability. The cloth shall show no appreciable distortion or puckering, and no dimensional change greater than 3.0 percent in the direction of the warp, 3.0 percent in the diagonal directions, and 2.0 percent in the direction of the filling when tested as specified in 4.4.

3.5 Nonfibrous materials. The total chloroform-soluble and water-soluble material content of the finished cloth shall not exceed 1.0 percent when tested as specified in 4.4.

3.6 pH. The pH value of the finished cloth shall be no lower than 5.0 nor higher than 8.5 when tested as specified in 4.4.

3.7 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be in continuous pieces, each not less than 50 yards. Each piece shall be put up on a roll as specified in PPP-P-1133.

3.8 Face identification. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the piece.

3.9 Fiber identification. Each roll shall be labeled, ticketed or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

* 3.10 Workmanship. The finished cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the point level specified.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.2.2 Examination of the end item. Examination of the end item shall be in accordance with 4.2.2.1 through 4.2.2.3.

* 4.2.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects, as defined in Section III of TED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1. No linear yard, (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (rolls) 1/
3200 or less	8
3201 up to and including 10,000	13
10,001 and over	20

1/ No more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls in which case all shipping containers shall be present in the sample.

The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 25.0 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 38.0 points.

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If no individual roll exceeds 38.0 points per 100 square yards, the lot shall be acceptable with respect to this characteristic. If one roll exceeds 38.0 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 38.0 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Bowed or bias filling exceeding 2 inch deviation from normal alignment
Baggy, ridgy or wavy cloth
Objectionable odor
Width less than specified
Uneven weaving
Tight selvage
Tight warp end

4.2.2.2 Examination for length.

4.2.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

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4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.2.2.3 Examination for face identification and compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for the defects listed below. The lot shall be unacceptable if two or more of the following defects are present:

Face identification missing from either or both ends.
Face identification on wrong side.
Not labeled or ticketed in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

* 4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1133, to determine that packaging, packing and marking comply with the section 5 requirements.

* 4.4 Testing of the end item. Methods of testing specified in FED-STD-191, wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on the sample unit for test purposes as specified in the applicable test method. The sample unit, except for dimensional stability, shall be 1 yard full width of the finished cloth. The sample unit for the dimensional stability determination shall be 2 yards, full width of the finished cloth. The sample size (number of sample units) except for the dimensional stability test, shall be as shown in the following table. The sample size for the dimensional stability test shall be 3 units regardless of lot size. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified.

Lot size in yards	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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* TABLE II. Test methods

Characteristic	Requirement paragraph	Test method
Fiber Identification		
Polyester (type I)	3.1.1.1	1600 2/
Nylon (type II)	3.1.1.2	1530 2/
Luster	3.1.1.1 and 3.1.1.2	2/
Multifilament	3.1.1.1 and 3.1.1.2	Visual 2/
Weight	3.3	5041
Yarns per inch		
Warp	3.3	5050
Filling	3.3	5050
Breaking strength	3.3	5100
Warp	3.3	5100
Filling	3.3	5100
Weave	3.3.1	Visual 1/
Finish	3.4	2/
Dimensional stability		
Warp	3.4.1	4.4.1
Filling	3.4.1	4.4.1
Diagonal	3.4.1	4.4.1
Nonfibrous material	3.5	2611
pH	3.6	2811

1/ One determination per sample unit and results reported as "pass" or "fail".

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2/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

4.4.1 Test for dimensional stability. The test specimen shall be a square of cloth at least 20 by 20 inches. While laid flat without tension, it shall be inscribed with an 18 by 18 inch square approximately equidistant from the edges, and with sides parallel to the warp and normal filling directions. The marked specimen shall be placed in a preheated oven and maintained at a temperature of 350°F ± 20°F for 10 minutes. The specimen may be looped over a rod or laid flat on a rack but shall have free air circulation on both sides. The rod or rack shall be of low-conductivity (non-metallic) material. Following the heating period the specimen shall be removed promptly from the oven and allowed to cool to room temperature on a flat surface, and then brought to equilibrium under standard conditions as defined in FED-STD-191. Any specimen showing appreciable distortion (skewing or bowing) or puckering shall be cause for rejection of the lot. Each side and each diagonal of the inscribed square shall be measured, and the changes from the original warp, filling and the two diagonal dimensions shall each be reported as the average of the three specimens (six measurements) to the nearest 0.1 percent. Appreciable distortion shall be reported separately for each specimen tested. Appreciable, means a change that is immediately noticeable when comparing the tested specimen with the original. If closer inspection is required to make apparent a slight change, the change is not considered appreciable.

4.4.1.1 Number and location of specimens. Three specimens shall be tested from each sample unit. One specimen shall include the selvage on one side of the cloth, one the center area, and the third the selvage on the other side. Each specimen shall be from a different filling than the others.

* 5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2 and 6.2.1).

5.1.1 Levels A and C. The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2 and 6.2.1).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with PPP-P-1133.

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6. NOTES

6.1 Intended use. The cloths covered by this specification are intended for use as base materials for coated fabrics to be utilized in the manufacture of air supported shelters.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Width required (see 3.3.2).
(d) Minimum length, if other than specified (see 3.7).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts and orders, it must be noted that various options, choices and alternatives as indicated in PPP-P-1133, may be exercised in the preparation of this cloth for delivery.

6.3 A temperature of 390°F ± 10°F is recommended for heat treating (see 3.4).

* 6.4 The margins of this specification are marked with an asterisk to indicate where (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-43286B CLOTH, PLAIN WEAVE, NYLON; CLOTH, PLAIN WEAVE, POLYESTER

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A GIVE PARAGRAPH NUMBER AND WORDING

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